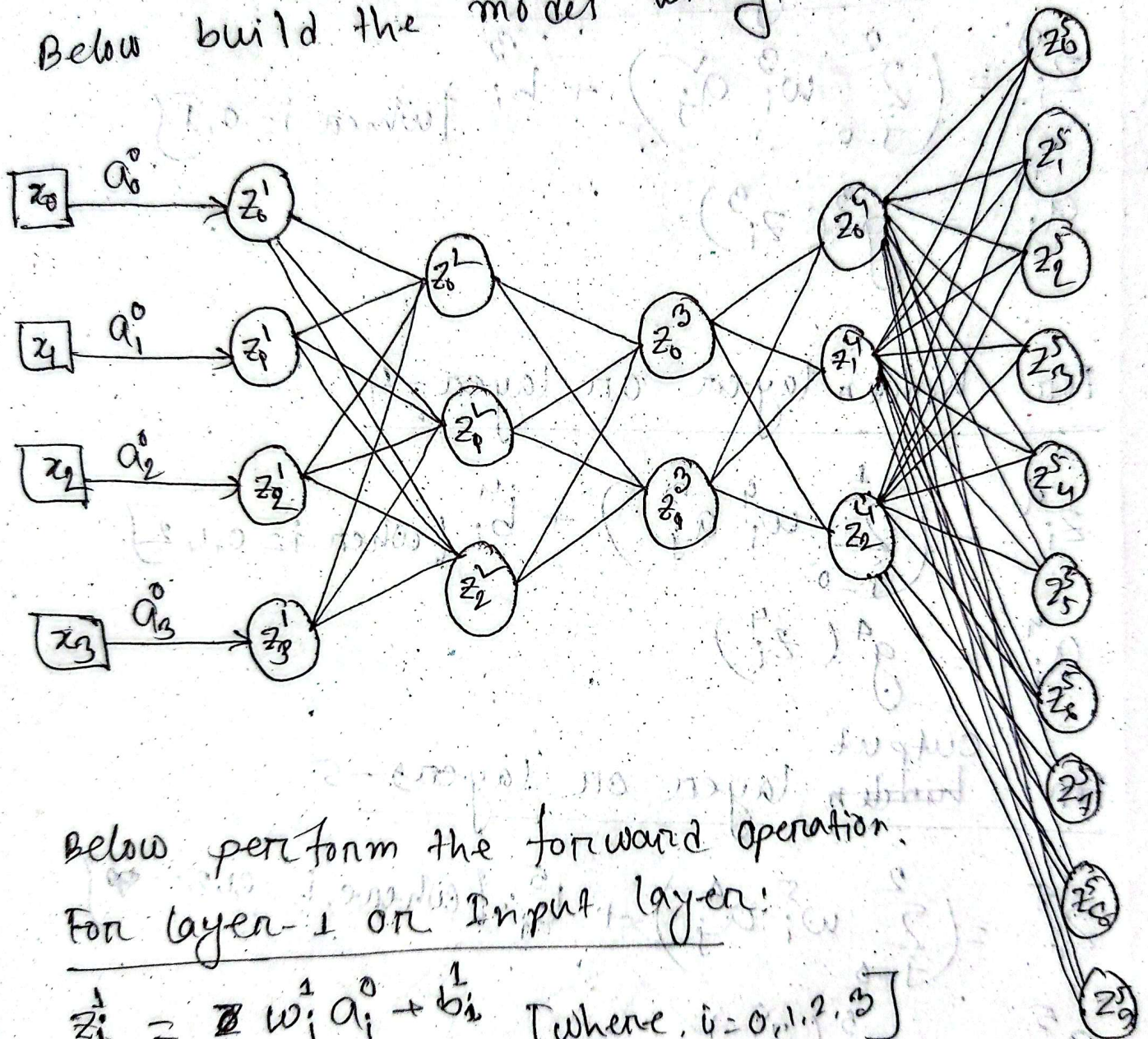


Assignment - 4
 ID: 2010676104
 Session: 2019-2020

Below build the model using 3 hidden layers



Below perform the forward operation.

For layer-1 or Input layer:

$$z_i^1 = \sum w_i^1 a_i^0 + b_i^1 \quad [\text{where } i=0,1,2,3]$$

$$a_i^1 = g^1(z_i^1)$$

For hidden layer or in layer-2

$$z_i^2 = \left(\sum_{j=0}^3 \tilde{w}_{ij} a_j^1 \right) + \tilde{b}_i \quad [\text{where } i=0,1,2]$$

$$a_i^2 = g^2(z_i^2)$$

For hidden layer on layer-3.

$$z_i^3 = \left(\sum_{j=0}^2 w_{ij}^3 a_j^2 \right) + b_i^3 \quad [\text{when } i = 0, 1]$$

$$a_i^3 = g^3(z_i^3)$$

For hidden layer on layer-4

$$z_i^4 = \left(\sum_{j=0}^3 w_{ij}^4 a_j^3 \right) + b_i^4 \quad [\text{when } i = 0, 1, 2]$$

$$a_i^4 = g^4(z_i^4)$$

For ^{output} ~~hidden~~ layer on layers-5

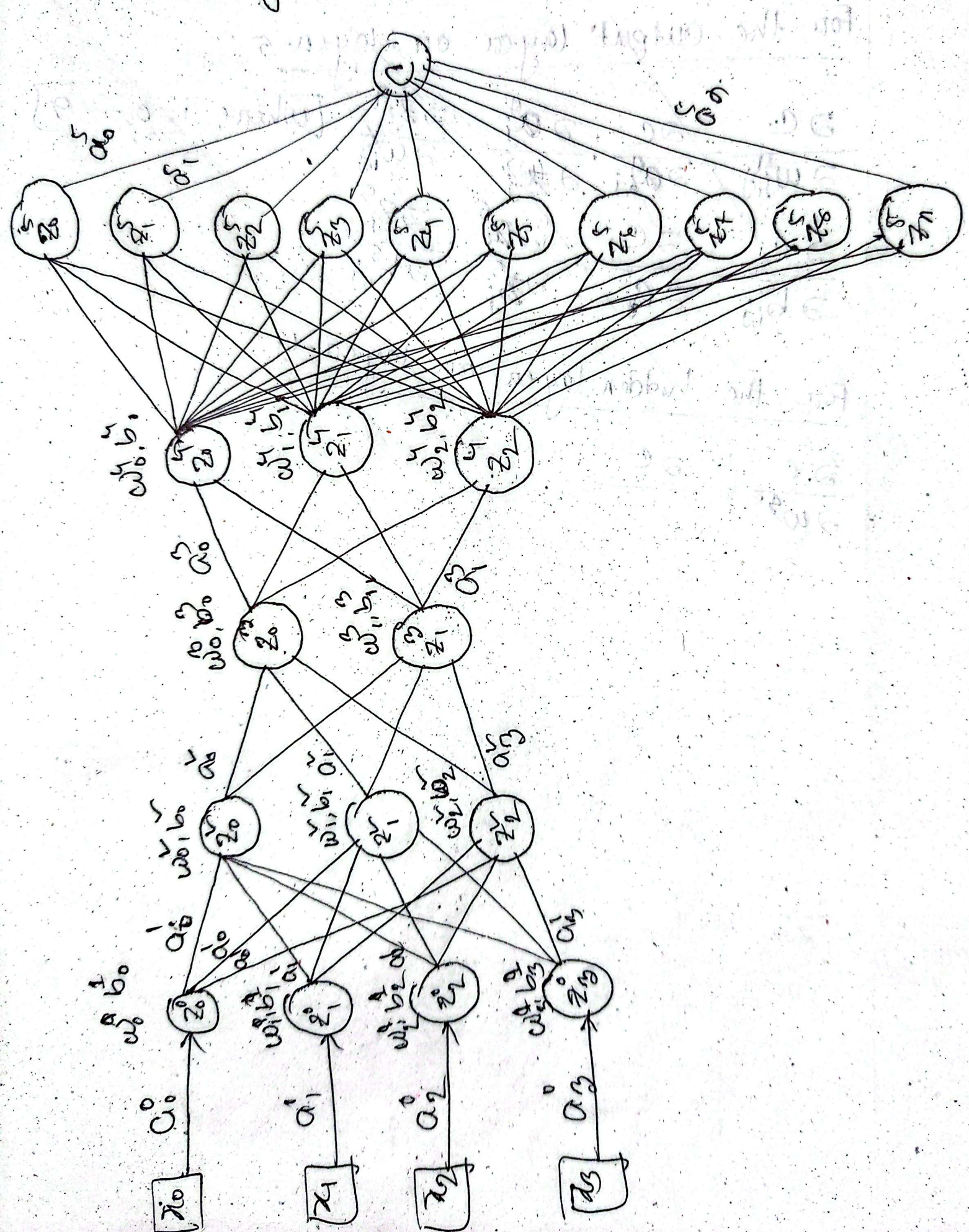
$$z_i^5 = \left(\sum_{j=0}^4 w_{ij}^5 a_j^4 \right) + b_i^5 \quad [\text{where, } i = 0, 1, \dots, 4]$$

$$a_i^5 = g^5(z_i^5)$$

Again,

$$C = \sum_{i=0}^9 \left[- (y_i \log a_i^5 + (1 - y_i) \log (1 - a_i^5)) \right]$$

Below given the forward propagation graph.



Below perform the back propagation.

For the layer-4

$$\frac{\partial C}{\partial w_{ij}^4} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial w_{ij}^4}$$

$$\frac{\partial C}{\partial b_{i,j}^4} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial b_{i,j}^4}$$

For the layer-3

$$\frac{\partial C}{\partial w_{ij}^3} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial w_{ij}^3}$$

$$\frac{\partial C}{\partial b_{i,j}^3} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial b_{i,j}^3}$$

For the layer-2

$$\frac{\partial C}{\partial w_{ij}^2} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial a_j^2} \cdot \frac{\partial a_j^2}{\partial z_j^2} \cdot \frac{\partial z_j^2}{\partial w_{ij}^2}$$

$$\frac{\partial C}{\partial b_{i,j}^2} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial a_j^2} \cdot \frac{\partial a_j^2}{\partial z_j^2} \cdot \frac{\partial z_j^2}{\partial b_{i,j}^2}$$

For the layer-1

$$\frac{\partial C}{\partial w_{ij}^1} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial a_j^2} \cdot \frac{\partial a_j^2}{\partial z_j^2} \cdot \frac{\partial z_j^2}{\partial a_j^1} \cdot \frac{\partial a_j^1}{\partial z_j^1} \cdot \frac{\partial z_j^1}{\partial w_{ij}^1}$$

$$\frac{\partial C}{\partial b_{i,j}^1} = \frac{\partial C}{\partial a_j^4} \cdot \frac{\partial a_j^4}{\partial z_j^4} \cdot \frac{\partial z_j^4}{\partial a_j^3} \cdot \frac{\partial a_j^3}{\partial z_j^3} \cdot \frac{\partial z_j^3}{\partial a_j^2} \cdot \frac{\partial a_j^2}{\partial z_j^2} \cdot \frac{\partial z_j^2}{\partial a_j^1} \cdot \frac{\partial a_j^1}{\partial z_j^1} \cdot \frac{\partial z_j^1}{\partial b_{i,j}^1}$$

Below given the back propagation graph

